OBJECT ORIENTED PROGRAMMING

[CT-260]

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DEPARTMENT: BCIT **BATCH:** 2025

YEAR & SECTION: FSCS-D

LAB :#12

QUESTION#1:

CODE:

```
#include <iostream>
#include <fstream>
using namespace std;

int main()
{
    fstream fout("./data.txt", ios::out | ios::trunc);
    if (!fout)
    {
        cout << "An error occured." << endl;
        return 1;
    }
    string input;
    cout << "Input data you want to store: ";
    getline(cin, input);
    cout << "Length: " << input.length() << endl;
    for (char c : input)
        fout.put(c);
    fout.close();
    ifstream fin("./data.txt");
    char ch;
    cout << endl << "\e[1;32mData retrived from the file.\e[33m" << endl;
    while (fin.get(ch))
        cout << ch;
    cout  << "\e[0m\n";
    return 0;
}
```

OUTPUT:

```
Input data you want to store: Hello, this is some data that will be saved on a
text file
Length: 58

Data retrived from the file.
Hello, this is some data that will be saved on a text file
```

Text file:

OOPspring26 [WSL: Ubuntu-24.04] - data.txt

```
Hello, this is some data that will be saved on a text file
```

QUESTION#2:

CODE:

```
#include <iostream>
#include <fstream>

using namespace std;

int main()
{
    ifstream sourceFile("input.txt", ios::in);
    if (!sourceFile)
    {
        cout << "Error: Cannot open the source file 'input.txt'!" << endl;
        return 1;
    }

    ofstream destFile("copy_of_input.txt", ios::out);
    if (!destFile)
    {
        cout << "Error: Cannot create or open the destination file!" << endl;
        sourceFile.close();
        return 1;
    }

    char ch;
    while (sourceFile.get(ch))
        destFile.put(ch);

    cout << "\e[1;32mFile copied successfully in text mode!\e[0m" << endl;
    sourceFile.close();
    destFile.close();
    return 0;
}
```

OUTPUT:

```
File copied successfully in text mode!
```

INPUT FILE:

OOPspring26 [WSL: Ubuntu-24.04] - input.txt

```
Here is some text
that will be copied
exactly into another
FILE :)
```

COPIED FILE:

OOPspring26 [WSL: Ubuntu-24.04] - copy_of_input.txt

```
Here is some text
that will be copied
exactly into another
FILE :)
```

QUESTION#3:

CODE:

```
#include <iostream>
#include <fstream>
#include <cstring>
using namespace std;

class Person
{
private:
    char name[25];
    int age;

public:
    Person() {}
    Person(char *name, int age) : age(age)
    {
        strncpy(this->name, name, 24);
        this->name[24] = '\0';
    }
    void print()
    {
        cout << name << endl
             << age << endl;
    }
};

int main()
{
    char name[25];
    int age;
    cout << "Enter your name: ";
    cin.getline(name, 24);
    cout << "Enter your age: ";
    cin >> age;
    Person person1(name, age);
    fstream file("./person.bin", ios::binary | ios::out | ios::trunc);
    if (!file)
    {
        cout << "\e[1;31mFile can't be generated.\e[0m\n";
        return 1;
    }
```

```
    file.write((char *)&person1, sizeof(person1));
    file.close();

    file.open("./person.bin", ios::binary | ios::in);
    if (!file)
    {
        cout << "\e[1;31mFile not available.\e[0m\n";
        return 1;
    }
    Person person2;
    if (file.read((char *)&person2, sizeof(person2)))
    {
        cout << "------\e[1;32mInfo retrived from the file\e[0m------\n";
        person2.print();
    }
    else
        cout << "\e[1;31mAn error occured while reading the file.\e[0m\n";
    return 0;
}
```

OUTPUT:

```
Enter your name: Taha Ahmed
Enter your age: 18
------Info retrived from the file------
Taha Ahmed
18
```

FILE (./person.bin):

QUESTION#4:

CODE:

```cpp
#include <cstring>
#include <iostream>
#include <fstream>
using namespace std;

class Participant
{
private:
    int ID, score;
    char name[25];

public:
    Participant() : ID(0), score(0)
    {
        name[0] = '\0';
    }
    Participant(int ID, char *name, int score) : ID(ID), score(score)
    {
        strncpy(this->name, name, 24);
        name[24] = '\0';
    }

    int getID() { return ID; }
    int getScore() { return score; }
    void print()
    {
        cout << "Name: " << name << endl;
        cout << "ID: " << ID << endl;
        cout << "Score: " << score << endl;
    }

    void input()
    {
        fstream file("./participant.dat", ios::app | ios::binary);
        if (!file)
        {
            cout << "\e[1;31mFile error.\e[0m\n";
            return;
        }
        file.write((char *)this, sizeof(Participant));
```

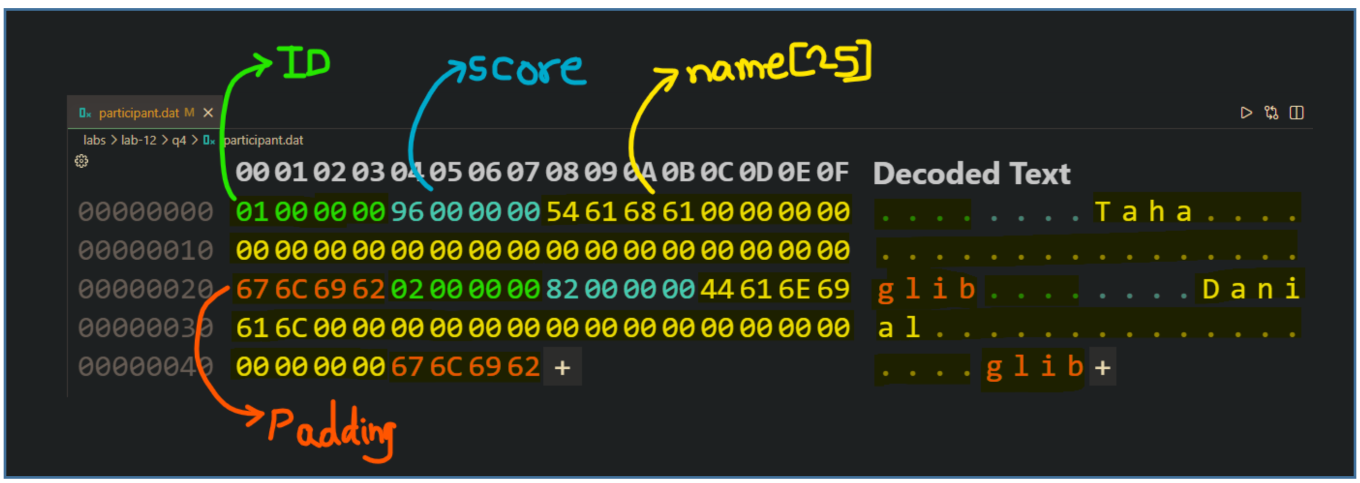
```cpp
    }

    void output(int IDin)
    {
        Participant p;
        fstream file("./participant.dat", ios::in);
        if (!file)
        {
            cout << "\e[1;31mNo Participants.\e[0m\n";
            return;
        }
        while (file.read((char *)&p, sizeof(p)))
        {
            if (p.getID() == IDin)
            {
                cout << "-----\e[1;32mTarget Found\e[0m-----\n";
                p.print();
                return;
            }
        }
        cout << "\e[1;31mID wasn't in the file.\e[0m" << endl;
    }

    void max()
    {
        Participant p, max;
        fstream file("./participant.dat", ios::in | ios::binary);
        if (!file)
        {
            cout << "\e[1;31mNo Participants.\e[0m\n";
            return;
        }
        while (file.read((char *)&p, sizeof(p)))
            if (p.getScore() > max.getScore())
                max = p;
        cout << endl
             << "\e[1;32mMaxima:\e[0m" << endl;
        max.print();
    }
};

int main()
{
    int ID, score;
    char name[25];
```

```cpp
    cout << "Enter your name: ";
    cin.getline(name, 24);
    cout << "Enter ID: ";
    cin >> ID;
    cout << "Enter score: ";
    cin >> score;

    Participant p(ID, name, score);

    p.input();

    cout << "Enter ID you want to search: ";
    cin >> ID;
    p.output(ID);

    p.max();

    return 0;
}
```

OUTPUT:

```
Enter your name: Danial
Enter ID: 2
Enter score: 130
Enter ID you want to search: 1
-----Target Found-----
Name: Taha
ID: 1
Score: 150

Maxima:
Name: Taha
ID: 1
Score: 150
```

FILE (./participants.dat):



QUESTION#5:

CODE:

```cpp
#include <iostream>
#include <fstream>
using namespace std;

int main()
{
    fstream file("./STORY.txt", ios::in);
    if (!file)
    {
        cout << "\e[1;31mFile not found.\e[0m\n";
        return 1;
    }
    string line;
    uint count = 0;
    while (getline(file, line))
        if (line[0] != 'A')
            count++;
    cout << "\e[1;32mTotal: " << count << "\e[0m\n";
    return 0;
}
```

OUTPUT:

```
Total: 3
```

FILE (./STORY.txt):

OOPspring26 [WSL: Ubuntu-24.04] - STORY.txt

```
The rose is red.
A girl is playing there.
There is a playground.
An airplane is in the sky.
Numbers are not allowed in the password.
```